Tips & Tricks has its own page numbers, and does not follow the rest of the magazine's page numbers. This is done to ensure that you can cut out and save T&T in a separate folder for later reference.

ROUTERS

Routers are often bought, set up and configured with the most basic settings asked for during the automated manual setup. However, they have a lot more to offer, which involves more than just letting you share your internet connection. Routers are devices powerful enough to enable wireless communication between all your devices. So why not take full advantage of their abilities and tweak them to get the most out of them.

Power up a router using a CAT5

Powering up a router over Ethernet works for wall mounted routers that don't have a plug point nearby for power supply. First of all, be ready to cut a DC power adapter plug compatible with your router.

Next, find the wires that are not in use on your CAT5 Ethernet cable. This cable has eight wires inside it, of which only four wires are used, while the other four are spare. These spare wires are generally connected to pins 4, 5, 7 and 8. Cut off the spare cables from the connector end and get rid of the insulation. Repeat this step at the other end of the ethernet cable as well. At one end of the ethernet cable, hook up the ends (4th with 5th and 7th with 8th) and join them to the connector's wires accordingly. Do the same with the other end of the cable with the power adapter's wires. Keep a check on the polarity and the connections you're making. Secure the open ends of the wires. This method works with approximately five meters of ethernet wire. Also, a Gigabit ethernet works on eight wires, so this trick might slow down the transfer speeds on that ethernet CAT5. If you need to use longer ethernet cables, you'd need to use a higher voltage power adapter, say, of 9V or 12V to power up a 5V router.

Try this trick to place your router where you want but can't due to lack of a plug!

Keep malware away from your network

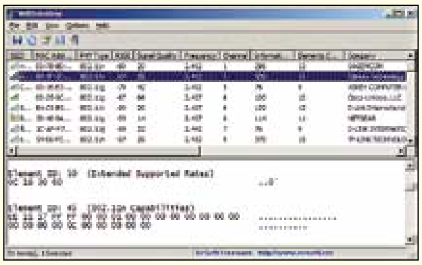
Routers and their network can get infected by malicious code if they have unsecured or exposed ports, which generally happens while using UPnP, remote management and other authentication tricks that cause the router to think it's dealing with a trusted code, when it's actually not! Infections often cause DNS and page redirections. 'History has it', 'The Moon', 'Moose', 'Chameleon' and other such worms and malware have been effectively evading your security protocols and infecting your router networks. It's difficult to get rid of them once they take over your network, so it's always a good idea to update the firmware, keep a check on all the security features and factory reset the router if it's been acting unusual. A very important security function involves turning off remote administration and limiting access to specific trusted IP addresses. Doing this can reduce the potential attack surface. Additionally, now and then check your Program Files folder for any unknown folders which may have been inserted by the malware. We recommend using network security essentials and anti-malware to safeguard your network from such attacks.

Let the 30-30-30 rule save your day!

Network fanatics have always been following the 30-30-30 rule, also called as the 'Hard Reset'. As the jargon implies, this is done by resetting any home used router to its default settings at any time.To perform a hard reset, hold down the device's reset button for 30 seconds while it's powered on. While still holding down the reset button, unplug the power cord from the router and hold for another 30 seconds. Holding down the reset button, power on the router again and hold for 30 more seconds. After this 30-30-30 second process is complete, your router should be restored to its factory default state. This is most efficient way to reset your router, however, never try it while your router firmware is upgrading or it might brick the router. Reserve this method for extreme cases; a soft reset should work in most other cases.

Choose your channel

Wi-Fi radio waves operate on channels that continuously interfere with other signals. This could be due to frequencies emitted by cordless phones, microwaves, other routers in vicinity, etc. Fortunately, channels 1, 6 and 11 are spaced far enough apart not to overlap or interfere with each other. So, on a non-MIMO setup (i.e. 802.11 a, b or g), you should always try to use channel 1, 6 or 11. If you want maximum throughput and minimal interference, channels 1, 6 and 11 are your best choice, however, depending on other wireless networks in your vicinity, one of these channels might be a much better choice than the others. There are tools that can help you find the clearest channel, such as 'Vistumbler', 'Xirrus Wi-Fi Inspector', 'Wifi-



Don't let your router operate on the most crowded channel!